Day 6 JavaScript Assignment - Higher Order Functions & DOM (Detailed)

Topics: map(), filter(), find(), reduce(), sort(), forEach(), Array of Objects, DOM Manipulation

Problem 1: Employee Management Dashboard

Business Scenario

Build an HR dashboard to display employee information and analytics.

Sample Input

Employees array of objects (id, name, department, salary, active).

Requirements

- Render employee cards using DOM.
- Filter IT employees.
- Filter active employees.
- Search by name using find().
- Increase salaries using map().
- Calculate total salary using reduce().
- Show summary statistics.

Expected Output

IT Employees: 2
Total Salary: ₹260000
Search Result: Priya

Learning Outcome

This assignment helps students practice Higher Order Functions on arrays of objects and dynamically update the webpage using the DOM.

Problem 2: Amazon Product Catalog

Business Scenario

Create an e-commerce product listing page with search and filtering.

Sample Input

Products array of objects (name, price, category, rating).

Requirements

- Render product cards.
- Filter by category.
- Search product.
- Sort by price.
- Calculate inventory value.
- Show average price.

Expected Output
Electronics Products Displayed
Average Price Calculated

Learning Outcome
This assignment helps students practice Higher Order Functions on arrays of objects and dynamically update the webpage using the DOM.

Problem 3: Student Result Portal

Business Scenario
Develop a student result dashboard for an LMS.

Sample Input
Students array of objects (roll, name, marks).

Requirements
- Render student table.
- Search by roll.
- Filter passed students.
- Generate grades.
- Calculate average.
- Highlight topper.

Expected Output
Average Marks: 72.75
Passed Students: 3

Learning Outcome
This assignment helps students practice Higher Order Functions on arrays of objects and dynamically update the webpage using the DOM.

Problem 4: Restaurant Order Management

Business Scenario
Create a restaurant order and billing dashboard.

Sample Input
Orders array of objects (item, price, quantity).

Requirements
- Render order cards.
- Calculate subtotal.
- Grand total using reduce().
- Filter expensive orders.
- Search order.
- Show quantity summary.

Expected Output
Grand Total: ₹1190
Orders Above ₹400 Displayed

Learning Outcome
This assignment helps students practice Higher Order Functions on arrays of objects and dynamically update the webpage using the DOM.

Problem 5: Movie Explorer

Business Scenario
Build a Netflix-style movie browsing application.

Sample Input
Movies array of objects (title, rating, genre).

Requirements
- Render movie cards.
- Search movie.
- Filter rating >=4.7.
- Display titles using map().
- Count movies.
- Sort by rating.

Expected Output
Trending Movies Listed
Total Movies: 4

Learning Outcome
This assignment helps students practice Higher Order Functions on arrays of objects and dynamically update the webpage using the DOM.

Submission Guidelines
- Each problem should be a separate project.
- Include index.html, style.css and script.js.
- Use Higher Order Functions instead of traditional loops wherever possible.
- Build the UI dynamically using DOM.
- Push code to GitHub and include screenshots in README.md.